AUGMENTED DECISION

From data to decisions

Losses, risks, and decision rules

Lecture 1 / Three hours, including activities and a break

Based on Lecture 1 of the original Integrated Course Syllabus.

All numerical cases in this lecture are constructed teaching examples.

A 30% rain forecast

Outdoor equipment will remain outside this afternoon.

- A forecast assigns probability 0.30 to rain.
- You can cover the equipment or leave it uncovered.
- Covering takes effort. Rain can damage uncovered equipment.

Which action would you choose?

What information is still missing?

Observations, predictions, and decisions

Observation	A forecast category, sensor reading, or recorded outcome.
Prediction	A statement about an unknown outcome, such as a rain probability.
Decision	An action we control, such as covering the equipment.

The same prediction can support different actions when consequences differ.

The objects of a decision problem

Symbol	Meaning	Equipment example
$a \in \mathcal{A}$	Action we choose	Cover or leave uncovered
$\theta \in \{0, 1\}$	State of the world	Dry or rainy
x	Available observation	A forecast category
$L(a, \theta)$	Loss after an outcome	Effort plus damage
$\delta(x)$	Decision rule	Action chosen from the forecast

Uppercase X and Θ denote random quantities. Lowercase x and θ denote their realized values.

Information available when the action is chosen

Before action	Observe the forecast. Check covers, staff, and time.
At decision	Choose an available action using that information.
After action	Observe the weather and record the resulting loss.

- A rule cannot use the afternoon's actual weather in the morning.
- An unavailable action must leave the feasible action set.

A valid formulation respects both timing and feasibility.

A loss table makes consequences explicit

Assume covering costs 2 loss units and prevents all rain damage.
Uncovered rain damage costs 10 loss units.

Action	Dry: $\theta = 0$	Rain: $\theta = 1$
Leave uncovered ($a = 0$)	0	10
Cover ($a = 1$)	2	2

Loss describes a consequence for each action-state pair.

These are stipulated teaching costs. Correct or protective actions may still have a nonzero cost.

Loss functions encode priorities

- Whose time, money, or harm enters the calculation?
- Can the consequences share a meaningful scale?
- Which requirements should remain hard constraints?
- Which costs are known, estimated, or disputed?

A forecast describes uncertainty about weather.
A loss table describes how we value the consequences.

Changing either can change the recommended action.

Exercise 1: an email filtering decision

An email is either legitimate or spam. You may deliver it or block it.

- Blocking a legitimate email costs 20 loss units.
- Delivering spam costs 4 loss units.
- The other two action-state pairs have zero loss.

1. Identify the actions and states, then write the 2×2 loss table.
2. Explain the difference between a false positive and a false negative here.
3. Name one additional input needed to choose an action for this email.

Exercise 1: the loss table

Action	Legitimate	Spam
Deliver	0	4
Block	20	0

- False positive: block a legitimate message. Loss $= 20$.
- False negative: deliver spam. Loss $= 4$.
- The table alone does not describe how likely each state is.

Counting all mistakes equally would discard the stated cost difference.

A probability model for uncertain states

Return to the equipment example. Let Θ denote the weather.

$$\mathbb{P}(\Theta = 1) = 0.30, \qquad \mathbb{P}(\Theta = 0) = 0.70.$$

- Probabilities are nonnegative and sum to one.
- They weight possible outcomes before the weather is known.
- An expected loss can differ from every individual realized loss.

We temporarily treat these probabilities as given. Lectures 2–4 address estimation and reliability.

Expected loss is a weighted average

For a fixed action a, define

$$\bar{L}(a) = \mathbb{E}[L(a, \Theta)] = \sum_{\theta} \mathbb{P}(\Theta = \theta) L(a, \theta).$$

1. List the possible states.
2. Multiply each loss by its state's probability.
3. Add the contributions and compare actions.

Choose an action with minimum expected loss under the stated model.

Background: UBC STAT 405, "Decision theory" [2].

Covering has the lower expected loss

$$\bar{L}(0) = 0.70 \cdot 0 + 0.30 \cdot 10 = 3,$$
$$\bar{L}(1) = 0.70 \cdot 2 + 0.30 \cdot 2 = 2.$$

Therefore $a^* = 1$: cover the equipment.

- Dry weather is more likely than rain.
- Covering still minimizes expected loss because damage is costly.

The most likely state and the best action answer different questions.

Different costs can reverse the decision

Keep the same 30% rain probability and damage cost of 10.

Cover cost	Leave: expected loss	Cover: expected loss	Choice
2	3	2	Cover
4	3	4	Leave

A recommendation needs its cost assumptions as well as its probabilities.

Exercise 2: when should we cover?

Covering again costs 2 in either state. Uncovered rain damage costs 10.

Rain probability	Loss if left	Loss if covered	Best action
$p = 0.10$	?	?	?
$p = 0.20$	?	?	?
$p = 0.60$	?	?	?

1. Complete the table.
2. Write one rule that works for every $p \in [0, 1]$.
3. Explain what happens at a tie.

Exercise 2: a simple decision rule

$$\bar{L}(0) = 10p, \qquad \bar{L}(1) = 2.$$

p	Leave	Cover	Choice
0.10	1	2	Leave
0.20	2	2	Either
0.60	6	2	Cover

One consistent convention is: cover when $p \geq 0.20$.

At $p = 0.20$, both actions minimize expected loss. A tie-breaking rule should be explicit.

Actions and decision rules

An action is the choice made today. A decision rule is a function:

$$\delta : \mathcal{X} \longrightarrow \mathcal{A}, \qquad a = \delta(x).$$

Suppose a forecast category is either high rain risk (H) or low (L).

$$\delta(H) = 1 \text{ (cover)}, \qquad \delta(L) = 0 \text{ (leave)}.$$

The rule specifies how available information changes the action.

Expected loss after observing information

After observing $X = x$, average over the conditional state probabilities:

$$\rho(a \mid x) = \mathbb{E}[L(a, \Theta) \mid X = x] = \sum_{\theta} L(a, \theta)\mathbb{P}(\Theta = \theta \mid X = x).$$

- x is known when we choose the action.
- Θ remains uncertain.
- The loss table stays the same unless the consequences also change.

$\rho(a \mid x)$ is conditional expected loss, also called posterior risk in a Bayesian model [2].

A Bayes action minimizes posterior expected loss

$$a^*(x) \in \arg\min_{a \in \mathcal{A}} \rho(a \mid x).$$

The corresponding Bayes decision rule chooses such an action for each x.

- The action set states what we can do.
- The conditional distribution describes the unknown state.
- The loss function measures the consequences.

Optimality is relative to these inputs. Errors in them can change the answer.

Background: UBC STAT 405, "Decision theory" [2].

A forecast can change the preferred action

Stipulate the following joint model for forecast and weather:

Forecast x	$\mathbb{P}(X = x)$	$\mathbb{P}(\text{rain} \mid x)$	$\rho(0 \mid x)$	$\rho(1 \mid x)$
High (H)	0.40	0.60	6	2
Low (L)	0.60	0.10	1	2

So $\delta^*(H) = 1$ and $\delta^*(L) = 0$.

$$\mathbb{P}(\text{rain}) = 0.40(0.60) + 0.60(0.10) = 0.30.$$

The model agrees with the original 30% rain probability before the forecast category is observed.

Risk evaluates the rule across cases

Under the specified joint model, define

$$R(\delta) = \mathbb{E}[L(\delta(X), \Theta)].$$

For the rule that covers only after H:

$$R(\delta^*) = 0.40 \cdot 2 + 0.60 \cdot 1 = 1.40.$$

Rule	Always leave	Always cover	Use forecast
Expected loss	3.00	2.00	1.40

Here risk averages over both observations and states. Appendix B distinguishes the fixed-state convention [3].

Loss, conditional expected loss, and risk

$L(a, \theta)$	Consequence for one action and one realized state.
$\rho(a \mid x)$	Average over unknown states after observing x.
$R(\delta)$	Average performance of a rule across cases.

Discuss: The rule leaves equipment uncovered after L, but rain occurs.
Does that outcome alone prove the rule was a bad decision?

A quick check before using data

A low forecast gives a rain probability of 0.10.

1. What is the expected loss of leaving equipment uncovered?
2. What is the expected loss of covering it?
3. If rain actually occurs after we leave it, what is the realized loss?

Then explain which numbers describe uncertainty and which describe consequences.

Empirical risk averages observed losses

Given n recorded cases (x_i, θ_i), calculate

$$\widehat{R}_n(\delta) = \frac{1}{n} \sum_{i=1}^{n} L\Big(\delta(x_i), \theta_i\Big).$$

- Apply the same rule to each case's available information.
- Score the chosen action against the recorded outcome.
- Average losses in consistent units.

An empirical average estimates performance on a population only under suitable sampling assumptions.

Exercise 3: three rules on six recorded days

Day	Forecast	Rain?	Always leave	Always cover	Cover only after *H*
1	*H*	Yes	?	?	?
2	*L*	No	?	?	?
3	*H*	No	?	?	?
4	*L*	Yes	?	?	?
5	*L*	No	?	?	?
6	*H*	Yes	?	?	?

1. Fill in each rule's realized losses and calculate its empirical risk.
2. Which rule wins on these six days?
3. Does that establish which rule will work best next month?

Synthetic observations. Cover costs 2; uncovered rain damage costs 10.

Exercise 3: the sample favors always covering

Day	Always leave	Always cover	Cover only after H
1	10	2	2
2	0	2	0
3	0	2	2
4	10	2	10
5	0	2	0
6	10	2	2
Total	30	12	16
Empirical risk	5.00	**2.00**	2.67

The earlier model gave the forecast rule a risk of 1.40.

A finite sample can differ from a model expectation.

Choosing a rule and evaluating it are separate jobs

If we choose among candidate rules using their sample losses,

$$\hat{\delta} \in \arg\min_{\delta \in \mathcal{D}} \widehat{R}_n(\delta),$$

we have performed empirical risk minimization over $\mathcal{D}$.

- Reusing that sample to judge the winner can favor a lucky rule.
- Separate evaluation cases provide a more honest check.
- Future conditions and the meaning of loss must remain relevant.

Lecture 2 develops model fitting, train-test splitting, and generalization.

Prediction methods also depend on a loss

Loss	Formula	What it emphasizes
Squared error	$(a-\theta)^2$	Large numerical errors
Absolute error	$\lvert a-\theta\rvert$	Error magnitude on its original scale
Zero-one error	$\mathbf{1}\{a \neq \theta\}$	Whether a label is wrong

A convenient prediction loss still needs a connection to the final decision.

Lab: how many rooms should a workshop reserve?

- Each room seats 25 people. At most three rooms are available.
- The workshop must reserve at least one room.
- Booking costs 80 units per room.
- Each attendee without a seat incurs a stated penalty of 10 units.
- There are 55 RSVPs. Actual attendance remains uncertain.

The objective is to minimize expected booking cost plus the shortage penalty.

The brief provides no attendance distribution.

Constructed classroom scenario. The penalty is a modeling assumption, not a market price.

A prompt that makes assumptions visible

Formulate the workshop decision using the brief. State the actions, unknown state, information available at booking, loss function, objective, and constraints.
Separate given facts from additional assumptions. Keep unknown probabilities symbolic.
Identify the evidence needed before recommending a number of rooms.

First write your own formulation. Then compare it with an LLM response or the supplied example.

A plausible-sounding response with hidden errors

> *Attendance is normally distributed with mean 55 and standard deviation 5. Reserve two rooms because* $55/25$ *is approximately 2. This gives a 95% chance that everyone has a seat. Booking cost is fixed, so it does not affect the optimum.*

Constructed flawed response for this exercise. This is not a recorded model output.

- Which claims lack support in the brief?
- Which mistakes can you detect without a probability calculation?

A defensible formulation can remain symbolic

Let $q \in \{1, 2, 3\}$ be rooms booked, D attendance, and x booking-time information.

$$L(q, D) = 80q + 10 \max(D - 25q, 0).$$

$$q^*(x) \in \arg\min_{q \in \{1,2,3\}} \mathbb{E}[L(q, D) \mid X = x].$$

- x includes the 55 RSVPs and any relevant past evidence.
- We still need a defensible model for attendance given x.
- We should confirm the shortage penalty and booking conditions.

Appendix D supplies an additional scenario model and a complete numerical example.

What belongs in the audit trail

Claim	What the response says and whether the brief supports it.
Check	A calculation, unit check, constraint check, or source of evidence.
Correction	The replacement statement or mathematical expression.
Open question	What remains unknown before taking action.

An LLM can help express a model. Responsibility for its assumptions and use stays with people.

A recommendation should expose its reasoning

A short decision memo should state:

1. The available action and the information used.
2. The probability and loss assumptions.
3. The comparison with a feasible alternative.
4. A limitation or condition that would change the choice.

Example: With rain probability 0.30, cover the equipment.
Expected loss is 2 versus 3 if left uncovered.
This assumes effective protection at cost 2 and damage cost 10.

Exit ticket

1. An email has spam probability 0.80. Blocking legitimate mail costs 20; delivering spam costs 4. Other losses are zero.
   Calculate both expected losses and choose an action.
2. In one sentence, distinguish a realized loss from empirical risk.
3. An LLM recommends two workshop rooms from the RSVP count alone.
   Name one missing input and one independent check.

Work individually. A calculator is allowed. Explain your reasoning briefly.

Exit ticket: worked answers

$$\mathbb{E}[L(\text{deliver}, \Theta)] = 0.80 \cdot 4 = 3.20,$$
$$\mathbb{E}[L(\text{block}, \Theta)] = 0.20 \cdot 20 = 4.00.$$

Deliver minimizes expected loss under these assumptions.

- Realized loss scores one action-outcome pair. Empirical risk averages recorded losses across cases.
- Missing input: an attendance distribution conditional on available information.
- Independent check: two rooms hold 50 people; compare the complete expected losses for all feasible room counts.

Sources and optional reading

1. **Course scope:** *Augmented Decision: Integrated Course Syllabus*, pp. 1–2, Lecture 1. Supplied course document.
2. **Expected loss and Bayes actions:** UBC STAT 405, "Decision theory." ubc-stat-ml.github.io / Decision theory
3. **Risk conventions:** Will Fithian, UC Berkeley, "Statistical decision theory" (2023). stat.berkeley.edu / Statistical decision theory

The equipment, email, workshop, and ordering examples are constructed teaching material. Their numerical assumptions are illustrative. The supplied flawed LLM response is also constructed.

Two averaging conventions for risk

Fixed-state risk holds θ fixed and averages over observations:

$$R_\theta(\delta) = \mathbb{E}_{X|\theta}[L(\delta(X), \theta)].$$

Bayes risk also averages over a prior π on the state:

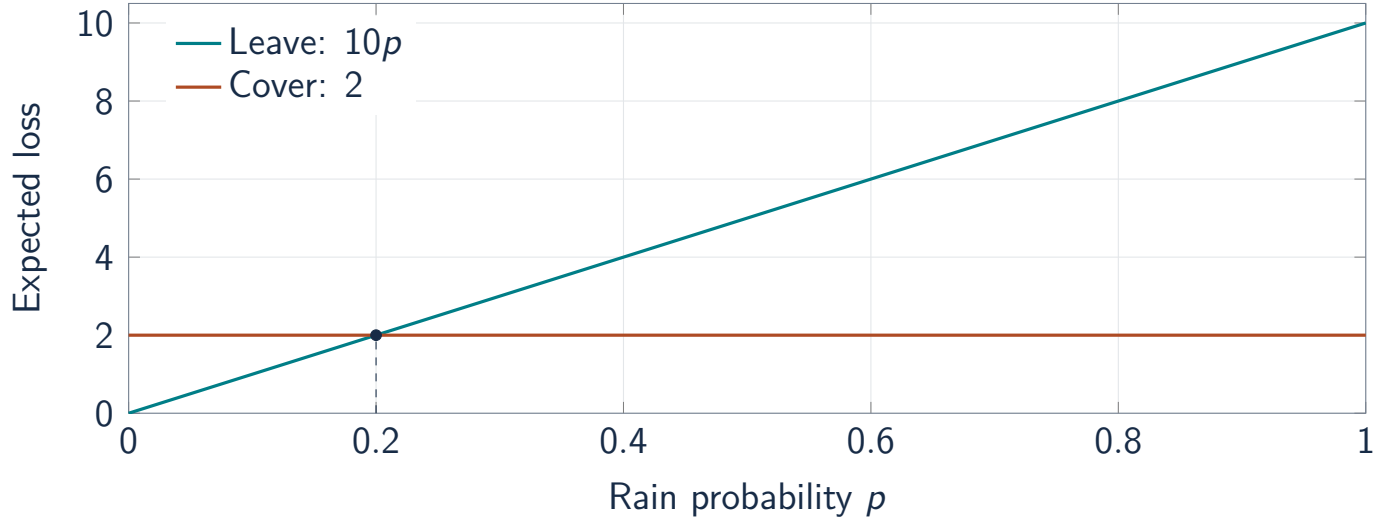
$$R_\pi(\delta) = \mathbb{E}_{\Theta\sim\pi}[R_\Theta(\delta)] = \mathbb{E}_{X,\Theta}[L(\delta(X), \Theta)].$$

A posterior-loss minimizing rule minimizes this joint average because

$$R_\pi(\delta) = \mathbb{E}_X[\rho(\delta(X) \mid X)].$$

The main lecture's $R(\delta)$ uses a specified joint model. Background: Fithian [3]. Finite states and actions suffice here.

The two expected-loss curves cross at $p = 0.20$



Workshop: an explicit attendance model

Now *add* these hypothetical probabilities, conditional on the 55 RSVPs:

$$\mathbb{P}(D = 40 \mid x) = 0.25, \quad \mathbb{P}(D = 55 \mid x) = 0.50, \quad \mathbb{P}(D = 70 \mid x) = 0.25.$$

Rooms q	Booking cost	Expected shortage penalty	Total
1	80	300	380
2	160	75	**235**
3	240	0	240

For two rooms, the expected shortage is

$$0.25(0) + 0.50(5) + 0.25(20) = 7.5 \text{ people.}$$

Two rooms minimize expected loss under this added model, by only 5 units.

Optional practice: a small ordering decision

A shop can order $q \in \{10, 20\}$ items for one day.

$$\mathbb{P}(D = 10) = 0.40, \qquad \mathbb{P}(D = 20) = 0.60.$$

Each leftover item costs 1 loss unit. Each unmet request costs 3.

1. Write $L(q, D)$ using positive parts.
2. Calculate expected loss for both quantities and choose one.
3. Repeat with leftover cost 5, keeping other assumptions fixed.
4. Name one assumption you would verify before using the answer.

Here $(z)_+ = \max(z, 0)$. These are incremental consequence costs, with no additional purchasing term.

Optional practice: worked solution

$$L(q, D) = (q - D)_+ + 3(D - q)_+.$$

$$\mathbb{E}[L(10, D)] = 0.40(0) + 0.60(30) = 18,$$
$$\mathbb{E}[L(20, D)] = 0.40(10) + 0.60(0) = 4.$$

Choose $q = 20$ for the original costs.

With leftover cost 5, the expected loss at $q = 20$ becomes 20.
The loss at $q = 10$ stays 18, so choose $q = 10$.

Useful checks include the demand probabilities, whether unmet demand is recorded, the allowed order quantities, and the meaning of the consequence costs.